

Coalition Game of Jammer Swarm for Combating Multifunction Radar

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Abstract—This paper deals with the autonomous task assignment problem of the cognitive jammer swarm for combating and degrading the multifunction radar system. The task assignment problem of jammers is modeled as a multiagent decision-making and control process, where jammers act as autonomous agents that can cooperatively make task assignment decision to pursue the best collective jamming utility, and a coalition formation game of jammers with transferable utility is established. To obtain the optimal control strategies of jammers, a multiagent coalition formation task assignment (MACFTA) algorithm is proposed. Numerical simulations are performed to verify the effectiveness of the proposed algorithm, and results illustrate that all jammers with the proposed strategy can work together concertedly to make the multifunction radar in low-threat working modes as much as possible.

Index Terms—Cognitive jammer swarm, multifunction radar, coalition game, task assignment, multiagent cooperation.

I. INTRODUCTION

In modern electronic warfare, one of the main tasks of the electronic jammer is to protect the high-value aircraft from being persistently and precisely detected by the enemy radar system [1]–[3]. The multifunction phased array radar possesses the capability of beam and waveform agility, especially with the development of the electronic counter-countermeasure technology [4]–[8], combating the multifunction radar to protect the high-value aircraft has become a major challenge for the electronic jammer.

In terms of space deployment or tactical application, the types of jamming include stand-off jamming, escort-support jamming, self-screening jamming, and stand-forward jamming [1]–[3]. Among them, stand-forward jamming is suitable for deployment on low-cost unmanned aerial vehicles (UAVs) [9]–[11], and it can be used to interfere with radars in the front, thereby protecting the high-value aircraft in the rear. However, the current stand-forward jamming is achieved through pre-deployment and pre-programming, where a large amount of jamming resources are indiscriminately deployed. It lacks the ability to adapt to dynamic environments, thus being unfavorable for resource conservation when combating the multifunction radar.

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Therefore, the adaptive or autonomous decision-making and control technology for the jammer swarm is urgently needed. Various multiagent cooperative control and optimization methods [12]–[14] have been widely studied and applied in relevant fields. In the field of electronic jamming, reinforcement learning [15]–[18] and evolutionary computation [19], [20] based decision-making and control methods have been presented in recent years. However, these results are applied for the scenario of the single jammer and cannot be directly applied to the jammer swarm.

To this end, this paper will explore the problem of autonomous cooperative task assignment for the jammer swarm to combat the multifunction radar, in which all the jammers act as autonomous agents, cooperatively making decisions to choose either jamming or silence tasks, with the aim of enabling the multifunction radar in a low-threat working mode with the least number of activate jammers. A solution based on coalition game theory [21] is, therefore, proposed to address this problem, and its effectiveness and superiority are verified by numerical simulations.

II. SCENARIO AND MODEL

A. Scenario

Consider a scenario where the UAV-enabled swarming jammers with escort for a high-value aircraft to penetrate a territory defended by a multifunction radar as depicted in Fig. 1. All jammers are denoted by set $\mathcal{N} = \{1, 2, \dots, N\}$, where N is the number of jammers. The multifunction radar can perform target search, target acquisition, target tracking, and missile guidance functions. Swarming jammers share information through wireless communication devices, and cooperatively make strategies to combat the multifunction radar to prevent the escorted target from being persistently and precisely detected.

Denote the set of radar functions as $\mathcal{Q} = \{1, 2, \dots, Q\}$. The multifunction radar can be transferred among these functions, but it can only operate in one of these functions in each time. For swarming jammers, the radar working in different functions causes distinct threats. The autonomous jammer swarm is considered, and it needs to allocate more resource to combat the radar operating in a high-threat function (e.g., missile guidance). The mission of the jamming swarm is to enable the radar operating in low-threat functions (e.g., target search) as possible with the least resources.

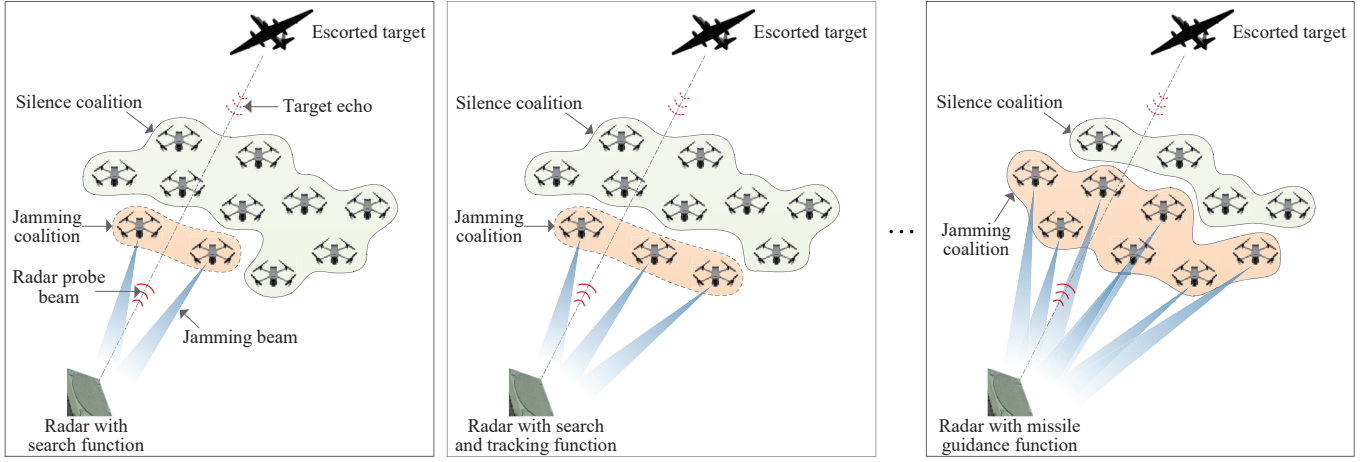


Fig. 1: The scenario of swarming jammers combating the radar with different functions.

B. Optimization Model

The mission of jammers is to assign jamming tasks to obtain the optimal collective utility. Thus, the control variable and the utility function are first designed, then the optimization model is established.

1) *Control Variable*: For task assignment, the task set is denoted by $\mathcal{T} = \{0, 1\}$, where task 0 represents the sleep task, that is, a jammer choosing task 0 indicates that it does not jam the radar, and task 1 represents the jamming task. The jammer only chooses to perform only one task at each time, and the task assignment variable of jammer $n \in \mathcal{N}$ for task $q \in \mathcal{T}$ at time k is given by

$$u_{n,q,k} = \begin{cases} 1, & \text{jammer } n \text{ chooses task } q \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

The task assignment variable of jammer n for all tasks at time k is given by $\mathbf{u}_{n,k} = [u_{n,0,k}, u_{n,1,k}]^T$, where $(\cdot)^T$ denotes the transpose of a vector or matrix. And the task assignment variable of all jammers for all tasks at time k is given by $\mathbf{u}_k = [\mathbf{u}_{1,k}^T, \mathbf{u}_{2,k}^T, \dots, \mathbf{u}_{N,k}^T]^T$.

2) *Utility Function*: The design criterion of the utility function is that the utility function can represent the revenue and cost of jammers through task allocation, and it is computable for jammers in practice.

The general form of the utility function of jammers for task assignment is given by

$$U(\mathbf{u}_k) = r(\mathbf{u}_k) - \lambda c(\mathbf{u}_k), \quad (2)$$

where $r(\cdot)$ denotes the reward function, $c(\cdot)$ the cost function, and $\lambda \geq 0$ the cost factor.

In this work, reward function $r(\cdot)$ is designed by

$$r(\mathbf{u}_k) = R(T(s_{k-1}, \mathbf{u}_k)), \quad (3)$$

where $s_k = T(s_{k-1}, \mathbf{u}_k)$ is the state (i.e., working function or working mode) of the radar, $T(\cdot)$ is the state transition function. The working mode determines the reward of the jammer swarm, which is affected by the jamming-to-signal ratio (JSR). Denote the JSR under control vector $\mathbf{u}_k$ as $\eta(\mathbf{u}_k)$,

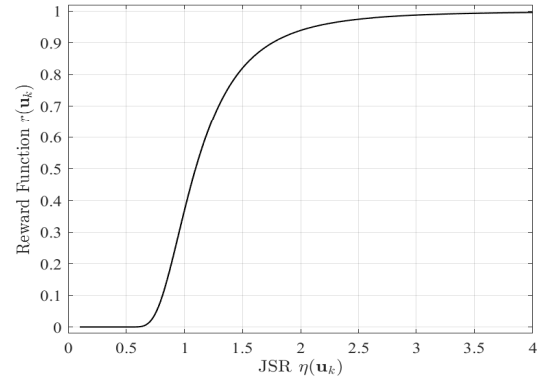


Fig. 2: The curve of the reward function versus the JSR.

then the reward of the jammer swarm at time time k can be set as follows

$$r(\mathbf{u}_k) = \exp\left(-\frac{1}{(\eta(\mathbf{u}_k))^L}\right), \quad (4)$$

where $L > 0$ is a constant. Let $L = 4$, the curve of the reward function versus the JSR is depicted in Fig. 2. It can be seen from Fig. 2 that the reward increases as $\eta(\mathbf{u}_k)$ increases, and the growth trend gradually becomes mitigatory as $\eta(\mathbf{u}_k)$ grows large, which satisfies the marginal utility theory [22].

Cost function $c(\cdot)$ is designed by

$$c(\mathbf{u}_k) = \sum_{n=1}^N u_{n,1,k}. \quad (5)$$

That is, the cost of the jammer swarm is the number of jammers that perform the jamming task. In practice, $c(\mathbf{u}_k)$ should be normalized.

3) *Optimization Model*: The optimization model is established by combining control variable and utility function as

$$\begin{aligned} & \max_{\mathbf{u}_k} U(\mathbf{u}_k) \\ & \text{s.t. } s_k = T(s_{k-1}, \mathbf{u}_k), \\ & \quad u_{n,q,k} \in \{0, 1\}, \\ & \quad \forall q \in \{0, 1\}, n \in \mathcal{N}. \end{aligned} \quad (6)$$

Note that this is a nonconvex optimization problem.

III. PROPOSED ALGORITHM

Since there is no control center to collectively make decision for all jammers, the autonomously multiagent cooperative decision-making method is required. As coalition game theory [21] can model and analyze rational agents for cooperative decision-making, this paper uses the coalition game to model cooperative task assignment behavior of jammers for combating the multifunction radar.

First, the coalition game of jammers is denoted as $\{\mathcal{P}, \mathcal{A}, v\}$, where $\mathcal{P} = \mathcal{N}$ represents the player set, $\mathcal{A} = \{0, 1\}$ the strategy set, and v the characteristic function (maximum utility function). From the perspective of coalition game theory, the optimization problem (6) is transferred to forming coalitions to obtain the maximum collective utility function. For task assignment of the jammer swarm, two coalitions are needed to be formed, i.e., jamming coalition $\mathcal{C}_{1,k}$ and silence coalition $\mathcal{C}_{0,k}$, where $\mathcal{C}_{1,k} \cap \mathcal{C}_{2,k} = \emptyset$ and $\mathcal{C}_{1,k} \cup \mathcal{C}_{2,k} = \mathcal{P}$. Jammers in $\mathcal{C}_{1,k}$ need to emit the jamming signal to jam the radar while jammers in $\mathcal{C}_{0,k}$ working in a stealth mode.

At each time k , jammers utilize previous knowledge (previous coalition partition $\mathcal{C}_{k-1}$ and radar state s_{k-1}) to form coalition partition $\mathcal{C}_k = \{\mathcal{C}_{1,k}, \mathcal{C}_{2,k}\}$. At time k , initial coalition partition $\mathcal{C}_{i,k} = \mathcal{C}_{i,k-1}$. For any jammer $n \in \mathcal{C}_{i,k}$ ($i \in \{0, 1\}$), it chooses to stay in the current coalition or join in another coalition $\mathcal{C}_{j,k}$ ($j \in \{0, 1\}$ and $j \neq i$) according to the following preference relation:

$$\mathcal{C}_{i,k} \succ_n \mathcal{C}_{j,k} \Leftrightarrow U_n(\mathcal{C}_{i,k}) > U_n(\mathcal{C}_{j,k}), \quad (7)$$

where $U_n(\mathcal{C}_{i,k})$ is the allocated utility of jammer n that is given by

$$U_n(\mathcal{C}_{i,k}) = \frac{U(\mathcal{C}_{i,k})}{N}, \quad (8)$$

with $U(\mathcal{C}_{i,k})$ being the utility of coalition $\mathcal{C}_{i,k}$. Since all jammers are cooperative completely, the utility of coalition $\mathcal{C}_{i,k}$ is the same as the utility of grand coalition $\mathcal{C}_k$, and the utility of grand coalition $\mathcal{C}_k$ is given by using the utility function (2) as

$$U(\mathcal{C}_k) = U(\mathbf{u}_k). \quad (9)$$

Here, coalition partition $\mathcal{C}_k$ is completely determined by control variable vector $\mathbf{u}_k$.

If preference relation $\mathcal{C}_{i,k} \succ_n \mathcal{C}_{j,k}$ holds, jammer n stays in current coalition $\mathcal{C}_{i,k}$, and coalition partition $\mathcal{C}_k$ keeps unchanged. Otherwise, jammer n leaves from current coalition $\mathcal{C}_{i,k}$ and join in another coalition $\mathcal{C}_{j,k}$, and coalition partition $\mathcal{C}_k$ is updated as follows

$$\mathcal{C}_k^{\text{new}} = \{\mathcal{C}_{i,k} \setminus \{n\}, \mathcal{C}_{j,k} \cup \{n\}\}. \quad (10)$$

All jammers make decisions in a predefined order, and the decision-making process is complete when the following defined Nash-stable coalition partition is obtained.

Definition 1 (Nash-stable coalition partition): Coalition partition $\mathcal{C}_k^* = \{\mathcal{C}_{1,k}^*, \mathcal{C}_{2,k}^*\}$ is a Nash-stable coalition partition

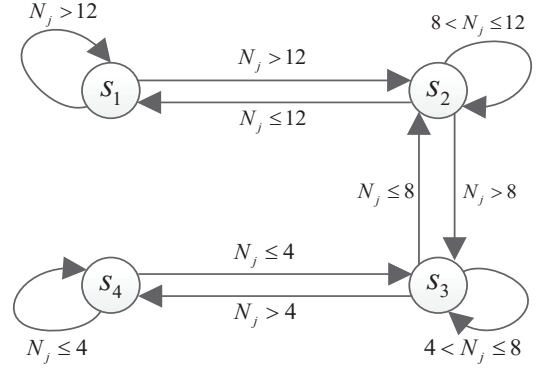


Fig. 3: The state transition of the multifunction radar.

if there is no another coalition partition $\mathcal{C}_k = \{\mathcal{C}_{1,k}, \mathcal{C}_{2,k}\}$ holds the following preference relation:

$$\mathcal{C}_{q,k} \succ_n \mathcal{C}_{q,k}^*, \forall n \in \mathcal{P}, \forall q \in \{0, 1\}. \quad (11)$$

At Nash-stable coalition partition, no jammer can obtain the improved utility by unilaterally changing its strategy.

IV. SIMULATION RESULTS

We consider the swarming jammers of the number of 20 to combat a multifunction radar, where the power of each jammer is $P_j = 5000\text{W}$, the power of the radar is $P_r = 50000\text{W}$, and the radar has four working states, i.e., target search (s_1), target acquisition (s_2), target tracking (s_3), and missile guidance (s_4). The transition of states is shown in Fig. 3, where N_j is the number of jammers choosing the jamming task. To verify the effectiveness of the proposed MACFTA algorithm, three counterparts are chosen for comparison: 1) Fixed task assignment (FixTA); 2) Random task assignment (RdmTA); 3) Maximum JSR task assignment (MJSRTA). Cost factor is set as $\lambda = 1.2$.

The number of jammers choosing the jamming task, the states of the radar, and the collective utility of the jammer swarm generated using FixTA, RdmTA, MJSRTA, and MACFTA are shown in Fig. 4, Fig. 5, and Fig. 6, respectively. As shown in Fig. 4, MJSRTA gives all the jammers to combat the radar since the largest number of jammers can achieve the maximum JSR, while MACFTA uses the jammers of the number of 14 to combat the radar since it needs to consider the cost of jammers caused by emitting signals. From Fig. 5, we see that both MJSRTA and MACFTA can enable the radar to operate in a low-threat target search mode, while FixTA and RdmTA would enable the radar to operate in the tracking mode with a higher threat level. In addition, it can be seen from Fig. 6 that the proposed MACFTA algorithm provides the maximum collective utility in all the compared methods. Therefore, the above simulation results demonstrate that the proposed MACFTA algorithm achieves the maximum collective utility with the least number of activate jammers, and enables the radar to operate in the low-threat target search mode as much as possible.

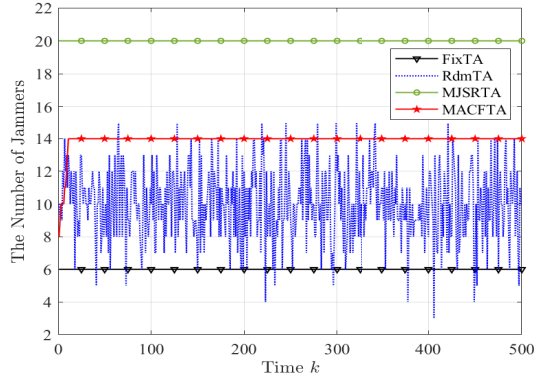


Fig. 4: The number of jammers choosing the jamming task by different task assignment methods.

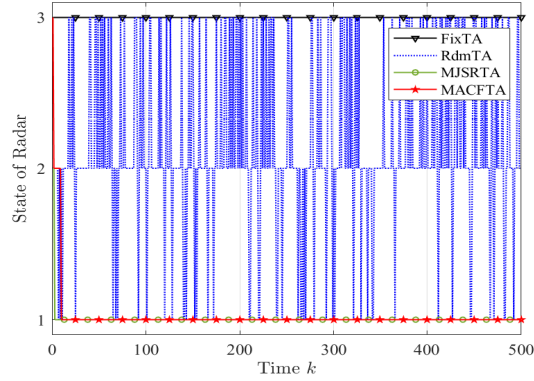


Fig. 5: The states of the radar by different task assignment methods.

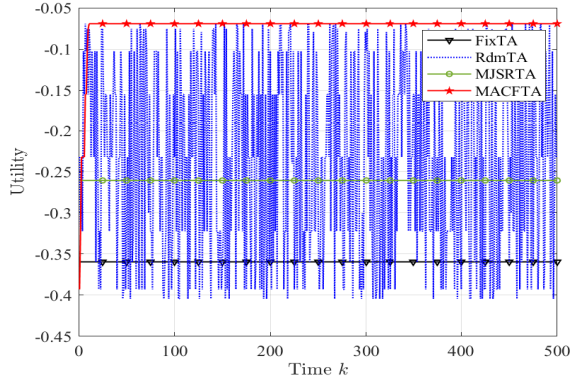


Fig. 6: The collective utility of the jammer swarm by different task assignment methods.

V. CONCLUSION

The autonomous task assignment problem of the jammer swarm for combating the multifunction radar is modeled as a multiagent cooperative decision-making process and formulated as a coalition game, in which each jammer acts as an autonomous agent to make jamming and silence decisions cooperating with other jammers to obtain the best collective utility. The MACFTA algorithm is proposed to seek the Nash-stable coalition partition, thereby obtaining the optimal cooperative task assignment strategy of the jammer swarm.

Numerical simulations illustrate that all the jammers using the proposed algorithm can autonomously make task assignment decisions cooperatively, ensuring that the multifunction radar operates in a low-threat mode while minimizing the number of activate jammers as much as possible.

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